

Sat, Sep 12 at 7:47 PM

Continue my KOHLER AI Research Lab case study. We selected Track 3 — Unified Enterprise AI Agent. Let’s start the research and project planning

Absolutely. We’ll continue with **Track 3 — Unified Enterprise AI Agent** and treat this as the beginning of the actual research + system-design phase.

One important finding already changes how I think we should approach the project: **KOHLER is not starting from zero with AI**. Microsoft reports that KOHLER has reached 98% Microsoft 365 Copilot adoption, has created **3,000 agents**, and is already exploring agentic AI.

So our project should **not** be “build a chatbot for KOHLER.” It should be:

Design a unified enterprise agent that can intelligently discover, retrieve, reason over, and act across KOHLER’s fragmented enterprise systems while respecting permissions, governance, and human approval.

That is a much stronger case-study direction.

1. What we know about KOHLER

KOHLER is a large global organization spanning Kitchen & Bath, Wellness and Hospitality, with a global workforce and manufacturing/operational footprint. Its 2024 report says KOHLER separated its Energy business, now Rehlko, and is focused on Kitchen & Bath, Wellness and Hospitality.

This matters because an enterprise agent at KOHLER potentially has to work across very different information domains:

Domain	Possible enterprise information
Product	Product specifications, manuals, catalogs
Engineering	Technical documentation, engineering knowledge
Manufacturing	Production data, plant information, quality

Domain	Possible enterprise information
Supply chain	Suppliers, inventory, logistics
Sales	Customers, opportunities, pricing
Marketing	Campaigns, brand assets, market information
HR	Policies, employee information
Finance	Budgets, reports, financial data
Hospitality	Resort operations, guest information
IT	Tickets, applications, infrastructure
Corporate	Policies, strategy, reports

The key problem therefore isn't simply knowledge retrieval.

It's enterprise orchestration.

2. The real problem we should solve

Imagine a KOHLER employee asks:

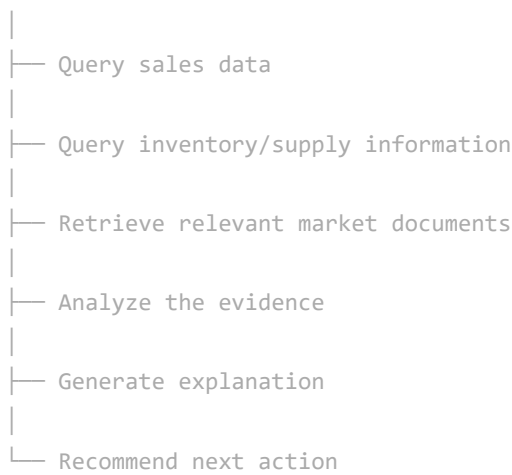
"Why did the sales of Product X decline in Europe last quarter, and what should we do about it?"

A normal RAG chatbot might retrieve documents about Product X.

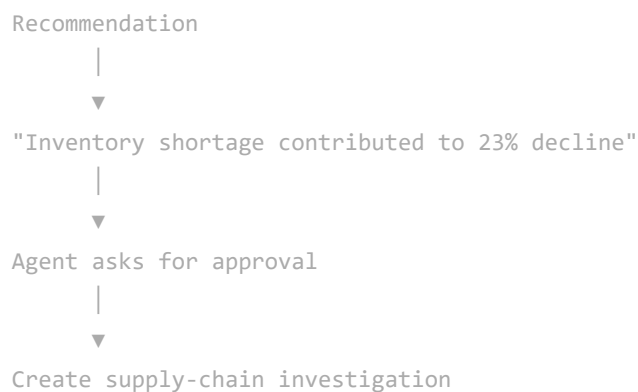
That's insufficient.

A true enterprise agent might need to:





And potentially:



That last part is what separates an **AI assistant** from an **agentic enterprise system**.

3. Our core project idea

I'm proposing that we frame Track 3 around:

KOHLER Enterprise Intelligence Agent

A unified AI agent that acts as an **intelligent front door** to KOHLER's enterprise **knowledge, data and business tools**.

The employee shouldn't need to know:

- which database contains the information
- which application owns the information
- which department owns the process
- which API should be called
- which document contains the answer

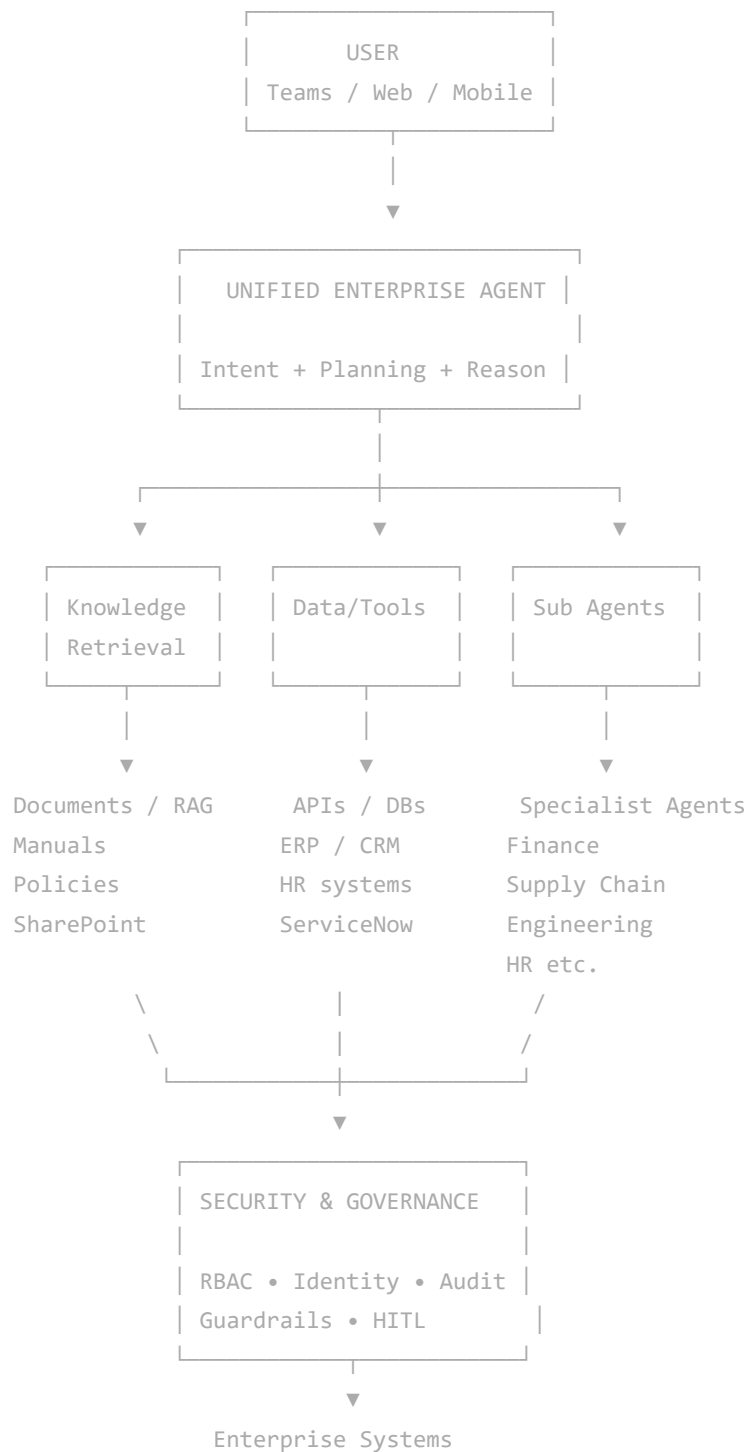
They simply ask:

"What do I need?"

The agent figures out the rest.

4. The architecture we should research toward

I would structure our eventual system into 6 major layers.



This architecture is strongly aligned with current enterprise-agent patterns.

Microsoft's current Foundry Agent Service, for example, provides managed agent runtime, tools, multiple model support, observability, identity/RBAC and governance capabilities.

5. The most important architectural distinction

I **do not** want us to build one giant agent with 30 tools.

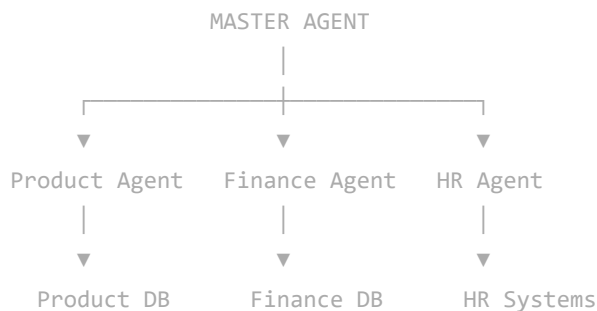
Instead:

Level 1 — Unified Agent

The user talks to one agent.

Level 2 — Specialist agents

The unified agent can delegate:



Potential specialists:

- Knowledge Agent
- Product Agent
- Sales Agent
- Supply Chain Agent
- Finance Agent
- HR Agent
- IT Agent

The master agent decides **which specialist is appropriate**.

This is where our agentic-AI knowledge will become useful.

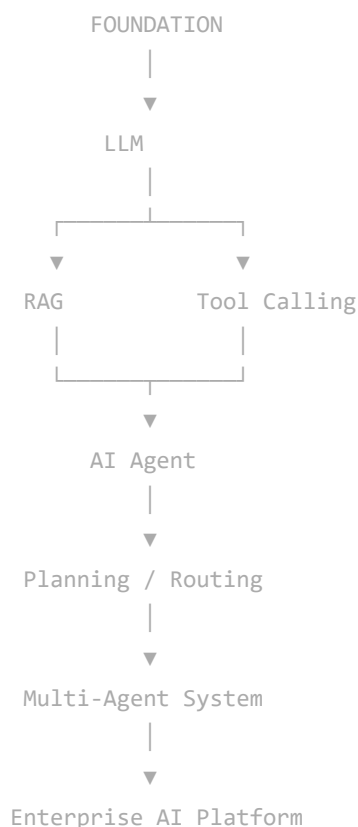
6. RAG is still important — but RAG isn't the whole project

Since you're planning to learn:

ML → DL → LangChain → RAG → Agentic AI

this project fits that progression very well.

Think of it as:



RAG answers:

"What does the company know?"

Tools answer:

"What can the system access/do?"

Agents answer:

"What should I do to accomplish this task?"

That distinction should become central to our case study.

7. Enterprise data architecture

We need to research two fundamentally different information paths.

A. Unstructured knowledge

Examples:

PDFs
DOCX
Policies
Manuals
Emails
Reports
Product documentation
Training material
Web pages

→ RAG / enterprise search

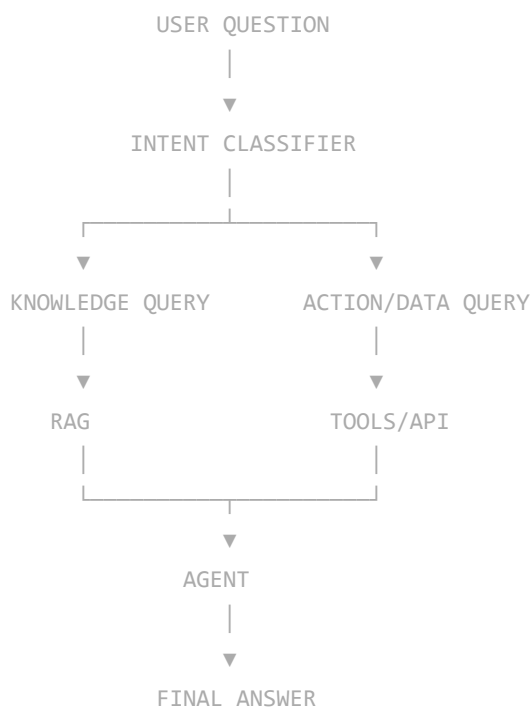
B. Structured / transactional data

Examples:

Sales
Inventory
Orders
Employees
Tickets
Financial metrics
Production numbers
CRM records

→ APIs / SQL / connectors / tools

So:



Modern Copilot Studio supports both connector-based enterprise knowledge and connectors as tools/actions, which reinforces this architecture.

8. Security has to be a first-class feature

This is probably one of the most important things we can put in the competition presentation.

Imagine:

Employee A asks:
"Show me the salary information for the engineering department."

The LLM shouldn't decide whether the employee is allowed to see it.

The system must enforce permissions **before the data reaches the model**.

So:



Microsoft's enterprise connector architecture explicitly emphasizes preserving source-level permissions, while current agent governance capabilities include identity, RBAC, data policies, connector controls and audit/observability mechanisms.

This gives us a powerful design principle:

The LLM reasons over authorized information; it does not determine authorization.

9. Human-in-the-loop

Another important component.

The agent should have **different autonomy levels**.

Level 0 — Answer

"What is our warranty policy?"
↓
Answer

Level 1 — Recommend

"Why are sales declining?"
↓
Analysis + recommendation

Level 2 — Prepare

"Contact the supplier about this issue."
↓
Draft email
↓
Ask approval

Level 3 — Execute

"Create a support ticket."
↓
Tool call
↓
Confirmation

Level 4 — Autonomous

Only for carefully approved low-risk workflows.

This gives us a clean:

Risk → Autonomy framework

Risk	Agent behavior
Low	Autonomous

Risk

Agent behavior

Medium

Execute + log

High

Human approval

Critical

Human-only

10. What makes our project different from "ChatGPT for employees"

This distinction will matter enormously in judging.

Weak proposal

"We will create an AI chatbot that answers employee questions using RAG."

✗ Too generic.

Better

"We will build an enterprise AI assistant that connects to multiple business systems."

Better, but still common.

Our proposal

"We propose a unified enterprise agent that dynamically plans tasks, routes them to specialized agents and enterprise tools, retrieves only permission-authorized information, reasons across heterogeneous sources, and executes approved actions with complete observability and human oversight."

That's much closer to an actual enterprise agent platform.

11. The research questions we now need to answer

This is where I suggest we structure the actual research.

Research Area 1 — KOHLER

We need to understand:

- KOHLER's business structure
- major business functions

- existing digital infrastructure
- existing AI initiatives
- current agent adoption
- likely enterprise pain points

We already found a very important piece of evidence: KOHLER reports **3,000 agents** and 98% Microsoft 365 Copilot adoption.

This means our solution needs to **unify an already growing agent ecosystem**, rather than replace it.

Research Area 2 — Enterprise AI Architecture

We need to compare:

- RAG
- Agentic RAG
- tool calling
- function calling
- planner/executor architectures
- multi-agent systems
- agent routing
- MCP
- semantic layers
- enterprise search

Research Area 3 — Enterprise Data

Research:

- structured vs unstructured data
- vector databases
- hybrid search
- metadata filtering
- semantic search
- SQL agents
- APIs
- enterprise connectors
- real-time vs indexed knowledge

Research Area 4 — Security

Research:

- RBAC
- ABAC
- OAuth
- identity propagation
- data isolation
- prompt injection
- indirect prompt injection
- tool abuse
- data exfiltration
- secrets management
- audit logging

Research Area 5 — Agentic Reasoning

We need to understand:

```
graph TD
    A[User request] --> B[Intent]
    B --> C[Plan]
    C --> D[Tool selection]
    D --> E[Execution]
    E --> F[Observation]
    F --> G[Correction]
    G --> H[Final result]
```

And eventually:

```
graph TD
    A[Plan] --> B[Execute]
    B --> C[Observe]
    C --> D[ ]
```



Evaluate



Re-plan



Research Area 6 — Evaluation

This is often missing from student projects.

We need measurable metrics such as:

Retrieval

- Recall@K
- Precision@K
- MRR
- NDCG

Generation

- groundedness
- faithfulness
- relevance

Agent

- task completion rate
- tool selection accuracy
- planning success
- recovery rate
- latency
- cost

Enterprise

- unauthorized-access rate
- human-approval rate
- audit completeness
- hallucination rate

This will make our project much more research-oriented.

12. Proposed project phases

I recommend that we **don't start coding yet**.

We'll go in this order:

Phase 1 — Enterprise research

Now

Understand KOHLER + enterprise AI landscape.

↓

Phase 2 — Problem definition

Identify the exact enterprise problem our system solves.

↓

Phase 3 — Use-case selection

Choose perhaps 3–5 representative workflows.

For example:

1. Enterprise knowledge search
2. Cross-system business analysis
3. IT/service request
4. Product/engineering investigation
5. Approval-based action

↓

Phase 4 — System architecture

Design:

- master agent
- specialist agents
- RAG
- tools
- databases
- APIs
- security
- memory
- observability

↓

Phase 5 — Technology selection

Only then decide:

- Python
- LangGraph/LangChain
- LLM
- vector DB
- SQL
- FastAPI
- frontend
- etc.

↓

Phase 6 — Prototype

Build a **small but convincing vertical slice**, not a fake enterprise platform.

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Phase 7 — Evaluation

Create test scenarios and measure the agent.

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Phase 8 — Case-study presentation

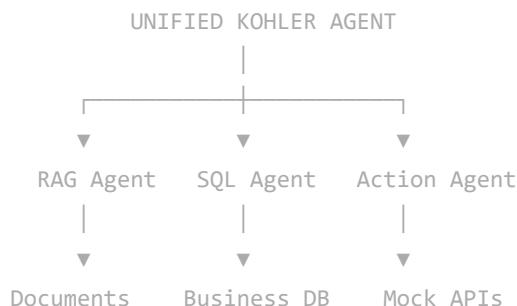
Turn the technical system into:

Problem → Research → Architecture → Prototype → Evaluation → Business Impact

13. Our initial MVP

For the actual prototype, I don't think we should attempt to connect 10 enterprise systems.

Instead:



Then demonstrate one complex question:

"Product X sales dropped in Europe. Investigate why and recommend what the sales team should do."

The agent could:

1. Query sales database.
2. Detect the decline.
3. Retrieve product documentation.
4. Retrieve inventory information.
5. Retrieve relevant market reports.
6. Compare evidence.
7. Produce an explanation.
8. Recommend actions.
9. Draft an action for human approval.

That single workflow could demonstrate **RAG + SQL + tools + planning + multi-source reasoning + agentic behavior + HITL**.

14. One thing I want us to avoid

Don't make the project:

"LangChain + vector DB + PDF chatbot."

That would underuse the Track 3 concept.

Our prototype should demonstrate:

Reason → Retrieve → Decide → Act → Verify

rather than:

Question → Search → Answer

15. Where we are now

Our working hypothesis is:

Problem

KOHLER has a rapidly expanding AI/agent ecosystem, but enterprise knowledge, data, applications and workflows remain distributed across systems.

Proposed solution

A **Unified Enterprise AI Agent** that serves as a single intelligent interface and orchestration layer across enterprise knowledge, data, specialist agents and business tools.

Core capabilities

Understand → Plan → Retrieve → Reason → Act → Verify

Core technical concepts

LLM + RAG + Tool Calling + Agent Routing + Multi-Agent Orchestration + Enterprise Search + RBAC + Human-in-the-Loop + Observability

Research objective

Determine the architecture that can make this system **useful, secure, explainable and realistically deployable** in an enterprise like KOHLER.

One particularly important finding from today's research is that KOHLER already has **3,000 agents**. That gives us a much stronger thesis:

Track 3 shouldn't be about building KOHLER's first agent. It should be about building the intelligence/orchestration layer that makes thousands of enterprise agents, knowledge sources and business tools accessible through one unified experience.

That is the direction I recommend we pursue.

Next, we should research KOHLER's actual enterprise ecosystem and existing AI initiatives in depth, then map those findings into a concrete problem statement and 3–5 high-value agent workflows.

Tuesday 12:33 AM

okay lets start with phase 1 in a new chat

Absolutely. Start a new chat and send:

"KOHLER AI Research Lab — Phase 1: Enterprise Research. Continue from Track 3 — Unified Enterprise AI Agent."

I'll treat that as the starting point and begin with KOHLER's business structure, existing technology/AI initiatives, enterprise ecosystem, current agentic-AI adoption, and potential pain points, using current sources and building the research systematically.